

Frozen Protocol

Frozen on 19 August 2026. *Tatiana Lvova*

After this date, no item changes except by a dated, written addendum stating the reason.

No addendum after the first test-window forecast is produced.

#	Step	Frozen Value
A	Sample	
A1	Dependent variable	daily count: opd + ipd, billed, russian-issued policies only
A2	Counting date	visit date
A3	Calendar	dense daily; no-case days = 0
A4	Extraction	20 August 2026 (confirmed billing completion for July 2026)
A5	Sample start	1 July 2023
A6	Sample end	31 July 2026
A7	Training sample	1 Jul 2023 - 30 Nov 2025 (884 observations)
A8	Test window	1 Dec 2025 - 31 Jul 2026 (243 observations)
B	Audit	
B1	Checks	missing dates, duplicates, impossible dates, negative counts
B2	Outlier policy	retain all genuine counts; no winsorising, smoothing or removal
B3	Corrections	documented structural errors only; log date, original value, corrected value, reason specified
C	Regressors	
C1-1	Calendar source	Consultant Plus production calendars 2023-2026 archive raw PDF files with download date
C1-2	Calendar encoding	xmlcalendar.ru JSON ru/2023-2026; archive raw JSON files with download date
C1-3	Reconciliation	every non-working day and every reverse transfer (weekend converted to a working day) verified against the PDFs
C1-4	Parse rule	non-working iff appears in month 'days' string without '**'; '**' = shortened working day; weekend absent from string = full working day
C1-5	Libraries	general holiday libraries not used (not aligned with production calendars)

C1-6	Inclusion rule	every uninterrupted run of ≥ 3 consecutive non-working days on the transferred calendar; no name-based selection (mechanism basis: Section 2.2); no merging across working days
C1-7	Generated set	17 clusters, written to holiday_clusters.csv, reproduced as Table 3; generated by script, not transcribed
C1-8	Grouping	H_NY = clusters containing 1 January (2 train / 1 test); H_OT = all remaining (9 train / 5 test)
C1-9	Pad	single (b, f) pair shared by both regressors
C1-10	Overlap handling	union, binary; overlapping padded windows never summed
C2-1	FX source	Bank of Russia official daily rate archive raw CSVs with download date
C2-2	FX nominal	N in published quote RUB per N THB; N = 10, verified against source
C2-3	FX variable	published quote, roubles per 10 baht; rescaled to roubles per 1 baht
C2-4	FX fill rule	value at date t = rate in force on t , i.e. the most recent determination effective on or before t .
C3-1	Search queries	“туры в Таиланд”, “туры в Тайланд”, “туры на Пхукет”, “туры в Паттайю”
C3-2	Search retrieval	Wordstat query-dynamics report: - Metric: number of queries - Frequency: weekly - Region: Russian Federation - Device type: all devices - Match: Wordstat default broad match - Download range begins no later than sample start minus the longest candidate lag
C3-3	Search index	four series summed; step-expanded, constant within its week
C3-4	Search vintage	week's value enters only from its publication date; record observed publication delay; archive raw CSVs with download date
D	Selection (training sample only)	
D1	Software	Python 3.12.13; statsmodels 0.14.6; pandas 2.3.3; numpy 2.5.2; scipy 1.18.0; pmdarima 2.1.1. Conda environment "thesis"; specification archived as environment.yml and requirements.txt in the project repository, tagged protocol-v1.0 at freeze. All runs logged in MLflow
D2	Differencing d	Candidate set d in $\{0, 1\}$: $d = 0$ only if ADF(Dickey & Fuller, 1979) rejects a unit root at 5% AND KPSS (Kwiatkowski et al., 1992) does not reject stationarity at 5%; otherwise $d = 1$. Tests applied to OLS residuals of y on a pre-declared pilot annual form (Fourier $K = 3$ plus day-of-week dummies), used for this test only. Declared restrictions relative to Hyndman & Khandakar (2008), which permits $0 \leq d \leq 2$ via repeated KPSS

		tests: $d = 2$ is excluded a priori, since it posits a stochastic trend in the growth rate, implausible for a bounded, seasonally recurring operational demand series, and would give forecast variance growing as h^3 at $h = 28$; and stationarity is required to be confirmed by two tests with opposite nulls rather than one. Confirmatory check: if $d = 1$, ADF and KPSS are re-run on the differenced residuals and the outcome reported; a failure is recorded as a diagnostic, not used to raise d
D3	Seasonal D	OCSB test at $s = 7$ (Osborn et al., 1988), as implemented in pmdarima; $D = 1$ if the test does not reject, else $D = 0$. Applies to the weekly period only: annual seasonality is deterministic by design and never differenced (Section 3.3)
D4	Constant	Included iff $d = D = 0$; forecasts otherwise follow a polynomial trend of order $d + D$ if a constant is retained (Hyndman & Athanasopoulos, 2021). Declared restriction: stricter than the Hyndman & Khandakar (2008) implementation convention, which permits a drift term at $d + D = 1$; no deterministic drift is imposed on bounded operational counts.
D5	Annual form and SARIMA orders (joint selection)	Candidates for the annual component: monthly dummies, and Fourier terms with $K = 1 \dots 6$ at period 365.25; when a constant is present, monthly dummies enter as eleven indicators with a reference month. For each candidate, the stepwise search of Hyndman & Khandakar (2008) is run with d and D fixed by D2-D3 and the weekly seasonal period $s = 7$ retained, and the resulting AICc recorded. The selected specification is the lowest-AICc combination of annual form and orders whose residuals pass the Ljung-Box test at lag $L = \max(2s, df + 3) = 14$ for $s = 7$ and typical df, tested at 5% with the chi-squared degrees of freedom reduced by the number of estimated ARMA parameters (statsmodels: model_df passed explicitly); lag rule per Hyndman & Athanasopoulos (2021) capped at $T/5$, which does not bind at $T = 884$. Procedure follows Hyndman & Athanasopoulos (2021), where the number of Fourier terms is selected by AICc with orders re-selected for each candidate. Declared restriction: K is capped at 6, since AICc tends to over-select Fourier terms at high seasonal periods and the training sample spans fewer than two and a half annual cycles. All candidates share d, D and the estimation sample, so the AICc values are comparable.
D6	Order fallback	if no candidate visited passes, take the stepwise winner and record the failure
D7	Holiday pad (b, f)	AICc over the 9 pads in the declared set $\{2,3,4\} \times \{2,3,4\}$; M1 orders and annual form held fixed
D8	FX lag	two-stage rule, declared: candidate lags 28-90 days; five largest CCF on training sample proceed to AICc; tie to shorter lag. CCF identification per Box et al. (2016)

D9	Search lag	candidate lags 28, 35, ..., 119 (multiples of 7); same five-candidate CCF then AICc rule
D10	M5 inheritance	inherits (b, f) from M2, FX lag from M3, search lag from M4; no re-tuning
D11	Scale trigger	Jarque-Bera rejects at 1% (Jarque & Bera, 1980) AND Spearman correlation between residuals and fitted values > 0.3; the threshold declared follows the conventional medium-effect benchmark (Cohen, 1988)
D12	Scale fallback	If D11 holds: $\log(y + 1)$ applied to the entire ladder. Point forecasts bias-adjusted on back-transformation to the count scale as $\exp(m_h + s_h^2 / 2) - 1$, where m_h and s_h^2 are the h-step forecast mean and variance on the log scale. Principle of bias adjustment follows Hyndman & Athanasopoulos (2021). However, the exact lognormal mean is used in place of the Taylor approximation given there, since Gaussian errors on the transformed scale are assumed by construction. Declared: bias-adjusted means are used throughout, consistent with the squared-error loss of the confirmatory tests, although MAE is minimized by the unadjusted median; the choice applies identically to every model in the ladder.
D13	Scale timing	decided once on training diagnostics, before test window; uniform across every model
E	Evaluation	
E1	Window type	expanding
E2	Origins	every 7 days from 30 Nov 2025 while origin + 28 ≤ sample end (31 origins)
E3	Re-estimated at each origin	coefficients only
E4	NOT re-selected	orders, annual form, pad, lags
E5	Horizons	$h = 7, 14, 28$
E6	Regressor availability	holiday regressors deterministic; both indicator lags ≥ 28 = longest horizon, therefore all values predate the origin
E7	Metrics	MAE (primary, reported first); RMSE; MASE scaled by M0 in-sample training MAE, computed over 1 Jul 2024 - 30 Nov 2025 (the training days having a year-earlier counterpart); MAPE/MdAPE computed on all days; a zero actual is treated as a data error and triggers re-verification against the source database
E8	Nested tests	Clark-West: M2, M3, M4, M5 vs M1; squared-error loss
E9	Non-nested test	modified Diebold-Mariano (HLN): M1 vs M0; squared-error loss

E10	Alternative	one-sided. Test statistics compared to t with n-1 dof, applying the small-sample convention of Harvey et al. (1997) to Clark-West, a conservative departure from the standard-normal reference of Clark & West (2007)
E11	HAC variance	Bartlett-kernel HAC (Newey & West, 1987), truncation $\text{ceil}(h/7)-1$; h-step errors follow MA(h-1) (Diebold & Mariano, 1995)
E12	Multiplicity	Holm {H1a, H1b, H1c} for each horizon, FWER 5%; horizons not mutually adjusted (Holm, 1979)
E13	H2	standalone at 5%, outside the Holm family
E14	Loss-function note	confirmatory decisions ride on Clark-West under squared-error loss; MAE remains primary descriptive metric; both reported
E15	Collinearity	VIFs reported for each augmented model, M2 through M5